

Highlights

UL-TransXNet: A Lightweight Architecture for Multi-Dataset Ultrasound Lesion Classification

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- UL-TransXNet classifies ultrasound lesions from ROI crops.
- Three public datasets support cross-organ evaluation.
- Ablations isolate MCA, MUDD, and differential attention.
- Automatic ROI experiments connect localization and classification.
- Android experiments quantify accuracy–latency trade-offs.

UL-TransXNet: A Lightweight Architecture for Multi-Dataset Ultrasound Lesion Classification

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Abstract

Ultrasound lesion classification is challenging because speckle noise, weak boundaries, low contrast, and acquisition-dependent variation obscure lesion cues. We present UL-TransXNet, a compact ROI-based TransXNet family architecture for benign-malignant ultrasound lesion classification. Lesion annotations are used to construct expanded square ROI crops, after which a lightweight hierarchical backbone models local texture and broader contextual relationships. We evaluate the model on three public datasets: TN5000 for thyroid nodules, BUSI for breast lesions, and AUL for liver lesions. Under the unified ROI protocol, UL-TransXNet obtains mean test AUC/balanced-accuracy pairs of 0.9483/0.8726 on TN5000, 0.8998/0.7961 on BUSI, and 0.8767/0.8200 on AUL. The result does not imply uniform superiority: ConvNeXt-Tiny remains slightly stronger on TN5000 at the selected operating point. Instead, the main finding is cross-dataset competitiveness, with the strongest average performance across the three datasets and clear gains on BUSI and AUL. TN5000 ablations show that the TransXNet fam-

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ily refinements improve operating-point behavior, while MCA contributes mainly to thresholded decision quality and cross-dataset robustness rather than to a uniform AUC gain. An auxiliary automatic-ROI experiment further shows that detector-predicted crops improve full-image classification on BUSI and AUL, with a remaining gap to oracle ROI performance. Android experiments quantify deployment trade-offs, where horizontal-flip test-time augmentation improves AUC from 0.9402 to 0.9466 at 226.2 ms per image. These results support UL-TransXNet as a competitive lightweight ultrasound classifier, while broader domain generalization remains open.

Keywords: Ultrasound image classification, Thyroid nodule, Breast ultrasound, Liver ultrasound, Lightweight neural network, Mobile deployment

1. Introduction

Ultrasound image analysis has become an important pattern recognition problem because ultrasound is widely used in screening, diagnosis support, and follow-up examination for multiple organs. Compared with computed tomography and magnetic resonance imaging, ultrasound is inexpensive, radiation-free, portable, and suitable for repeated acquisition. These advantages make ultrasound attractive for computer-aided diagnosis, but they also make the recognition problem visually difficult. Speckle noise, low contrast, blurred lesion boundaries, operator dependency, and device-specific appearance shifts can make benign–malignant discrimination substantially harder than standard natural-image classification. A useful ultrasound classifier therefore needs to preserve local lesion details, use surrounding tissue

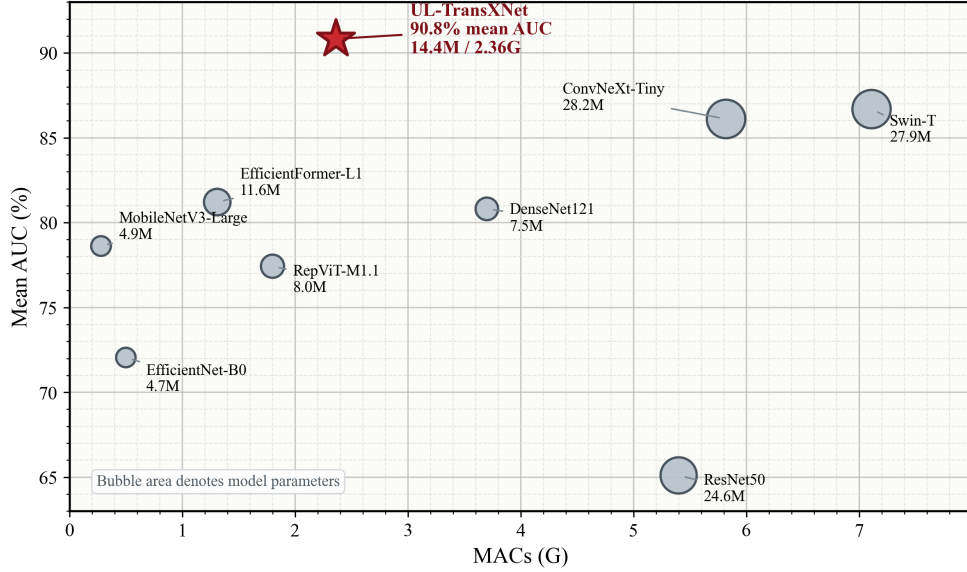


Figure 1: Accuracy-complexity trade-off on the three public ultrasound datasets. The vertical axis reports the mean AUC over TN5000, BUSI, and AUL; the horizontal axis reports MACs for one forward pass; bubble area denotes model parameters.

context, and remain compact enough for realistic deployment.

Public datasets now make it possible to evaluate ultrasound classifiers beyond a single organ. TN5000 provides a large public benchmark for thyroid nodule analysis [1]. BUSI is widely used for breast ultrasound lesion classification [2]. The Annotated Ultrasound Liver (AUL) dataset broadens the setting to liver ultrasound [3]. These datasets are not interchangeable: they differ in anatomy, acquisition appearance, class prevalence, annotation style, and split protocol. This makes them useful for testing whether an architectural idea is only tuned to one benchmark or remains competitive under different public ultrasound settings.

Recent deep learning methods have improved ultrasound analysis, but

two gaps remain important for practical lesion-classification systems. First, many studies report strong results on one dataset while providing limited evidence that the representation remains useful elsewhere. Second, increasingly complex backbones can obscure the practical trade-off between accuracy and deployability. This trade-off is not a secondary issue in ultrasound: many clinical support scenarios require stable inference under constrained hardware and repeatable preprocessing.

This paper addresses these gaps with UL-TransXNet, a compact TransXNet family classifier for ROI ultrasound lesions. The study is deliberately organized as an evidence chain rather than as a broad claim of universal superiority. TN5000 is treated as the primary large benchmark. BUSI and AUL test cross-organ competitiveness. Structure and module ablations quantify how the TransXNet-family refinements behave under the unified protocol. A mobile prototype then examines whether the resulting model family can occupy a practical accuracy–latency region.

The main contributions are summarized as follows:

- We present UL-TransXNet, a compact ROI-based architecture for ultrasound lesion classification that combines hierarchical local–global feature modeling with a simple lesion-centered input protocol.
- We evaluate the model on TN5000, BUSI, and AUL, and we explicitly separate cross-dataset competitiveness from stronger but unsupported claims of uniform superiority.
- We provide unified-protocol TN5000 ablations that distinguish the effects of the structural progression, MCA, MUDD, and differential attention.

- We report Android deployment measurements for a TN5000-calibrated prototype to show the practical speed-accuracy trade-off of the lightweight model family.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the proposed method. Section 4 presents the datasets and protocol. Section 5 reports the main experimental findings. Section 6 discusses the implications and limitations, and Section 7 concludes the paper.

2. Related Work

2.1. Ultrasound lesion classification

Ultrasound lesion classification has been studied most extensively in breast and thyroid imaging, where public data availability has enabled repeated benchmarking [2, 1]. Earlier breast ultrasound work often relied on handcrafted morphological and texture descriptors combined with conventional classifiers [4]. More recent deep models have shifted the focus toward representation learning directly from lesion-centered images or regions of interest. For example, Luo *et al.* proposed a multi-view learning strategy for breast ultrasound tumor classification in *Pattern Recognition*, showing that task-specific view construction can improve classification accuracy [5]. These studies confirm that ultrasound lesion classification benefits from architectures that can retain subtle boundary, texture, and contextual cues. However, the majority of published systems remain strongly tied to a single organ or a single benchmark.

Compared with breast ultrasound, public thyroid and liver ultrasound classification benchmarks are less mature but increasingly important. TN5000 has substantially improved the benchmarking environment for thyroid nodule analysis by providing a large-scale public dataset and standardized evaluation protocol [1]. The AUL dataset broadens public benchmarking to liver ultrasound by offering annotated benign, malignant, and normal cases [3]. Taken together, these datasets suggest that a useful ultrasound classification architecture should not only perform well on one organ, but also remain competitive when transferred to datasets with different anatomy, lesion appearance, and acquisition characteristics.

2.2. Efficient visual architectures

The design of efficient visual backbones has rapidly evolved from classical convolutional networks to compact CNN-Transformer hybrids. ResNet, DenseNet, EfficientNet, and MobileNetV3 established widely used convolutional design principles such as residual learning, dense feature reuse, compound scaling, and hardware-aware mobile optimization [6, 7, 8, 9]. Vision Transformer and DeiT then demonstrated the strength of attention-based image recognition, while Swin Transformer and ConvNeXt clarified the practical value of hierarchical representations and convolutional inductive biases [10, 11, 12, 13]. These models form the main reference space for modern visual backbones.

Recent lightweight designs focus less on raw parameter reduction and more on where computation should be allocated. MobileViT, EdgeNeXt, EfficientFormer, RepViT, and TransXNet are representative examples of this direction, combining local detail modeling, global context aggregation, and

deployment-aware efficiency in different ways [14, 15, 16, 17, 18]. This line of work is particularly relevant to ultrasound analysis, where lesions often occupy a small portion of the image and subtle structural changes can be diagnostically important. Our architecture is built in this efficient-backbone context rather than as a detached medical-only network.

Medical image analysis has also begun to adopt lightweight hybrid designs. LightCM-PNet emphasizes real-time ultrasound segmentation under deployment constraints [19], while Lite-MixedNet explores efficient CNN–Transformer fusion for medical image segmentation [20]. These works show that lightweight design is becoming a first-class consideration rather than a secondary implementation detail. Our work follows this trend, but targets ultrasound lesion classification rather than segmentation. The goal is not to propose an entirely separate paradigm, but to derive a coherent and task-motivated architecture that preserves efficiency while improving fine-grained lesion discrimination.

2.3. Multi-dataset validation in medical image analysis

Reporting results on a single benchmark is often insufficient in medical image analysis because performance may depend strongly on dataset-specific factors such as annotation protocol, class prevalence, acquisition device, and institution-specific image characteristics [21, 22]. This issue is especially pronounced for ultrasound, where both image quality and lesion appearance can vary substantially across clinical settings. Recent medical-image-analysis studies have therefore begun to supplement their main benchmarks with robustness analysis, external validation, or deployment-oriented testing [23, 19]. Such evidence does not automatically prove universal generalization,

but it gives a more credible picture of where a method is strong and where it remains fragile.

Our paper adopts this perspective. The primary claim is strong in-domain performance across several public ultrasound datasets, with TN5000 as the main benchmark and BUSI/AUL as cross-organ validation datasets. BUSI and AUL are retained as cross-organ validation datasets because they help characterize performance under different anatomy, lesion appearance, and acquisition characteristics. Temperature scaling is used only as validation-derived post-hoc calibration, following the established calibration literature [24]. This positioning keeps the evidence chain aligned with what the experiments can defensibly support.

3. Method

3.1. Problem formulation

Given an ultrasound region-of-interest (ROI) image x , the goal is to predict the lesion label $y \in \{0, 1\}$ for benign–malignant classification, where 0 denotes benign and 1 denotes malignant. The model outputs a two-dimensional logit vector $z = f_{\theta}(x)$, and the malignant posterior is computed as

$$p(y = 1 \mid x) = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)}. \quad (1)$$

The main paper uses the same binary formulation across TN5000, BUSI, and AUL, so that the proposed architecture can be evaluated under a unified lesion-classification setting.

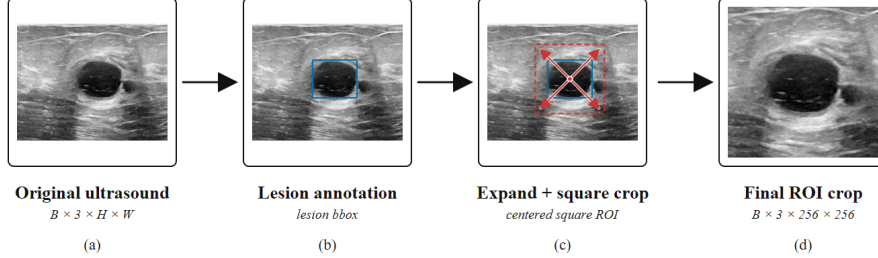


Figure 2: ROI-based input construction. The lesion annotation defines the initial bounding box; an expanded square crop centered on the lesion is extracted and resized to the model input size.

3.2. ROI-based input construction

Figure 2 summarizes the input construction. The lesion annotation defines the initial bounding box. The box is expanded around the lesion center, converted to a square crop, clipped to the image boundary, and resized to the model input size. This procedure keeps the lesion centered while retaining a controlled amount of surrounding tissue context. The main classification protocol therefore uses annotation-derived ROI crops; a separate closed-loop experiment in Section 4 evaluates detector-predicted ROI crops as an auxiliary inference-time front end.

3.3. Overall architecture

Figure 3 illustrates UL-TransXNet, the architecture used in the main experiments. A 256×256 ROI crop is processed by a lightweight four-stage hierarchical backbone. The backbone follows the tiny TransXNet family configuration with stage depths $[4, 4, 15, 4]$, embedding dimensions $[48, 96, 224, 448]$, head numbers $[1, 2, 4, 8]$, and spatial-reduction ratios $[8, 4, 2, 1]$. Each block

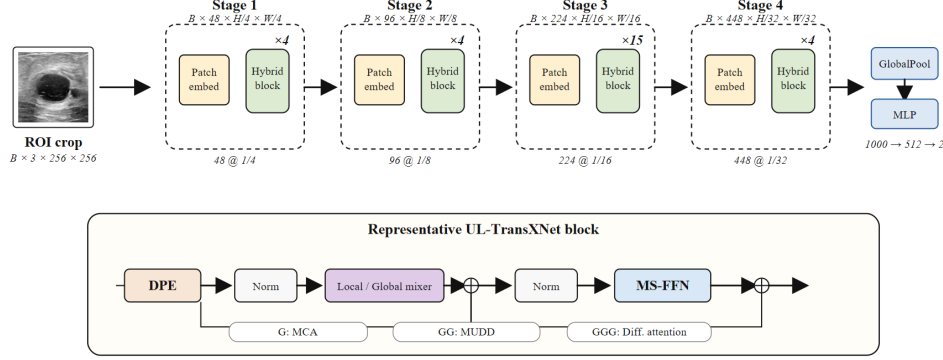


Figure 3: Overview of UL-TransXNet. The ROI crop is resized to 256×256 and processed by hierarchical patch/stem embedding, four TransXNet stages, global pooling, and a lightweight MLP head. The lower panel abstracts the representative block used to describe local/global token mixing, residual fusion, and MS-FFN refinement.

keeps the central TransXNet idea of local–global token mixing: local convolutional dynamics preserve fine lesion structure, while the global branch models broader contextual relationships.

The backbone is used as a feature extractor and outputs a 1000-D representation. A lightweight task head maps this representation to the final binary prediction through a two-layer multilayer perceptron, $1000 \rightarrow 512 \rightarrow 2$, with GELU activation and dropout between the two linear layers. This keeps the classifier head simple and makes the backbone design the main source of representation capacity.

3.4. Progressive TransXNet family refinements

Rather than treating the final model as a one-shot design, we organize the architecture as a progressive refinement of the TransXNet baseline [18]. The

baseline, denoted **TransXNet**, already combines dynamic local convolution and global attention in a compact hierarchical backbone. The ultrasound setting motivates additional refinements because lesion boundaries can be weak, discriminative textures can be local, and useful context can appear at different spatial scales.

The first refinement, **TransXNet-G**, enables multi-scale coordinate attention (MCA), inspired by coordinate attention for lightweight networks [25], in selected global-branch locations. The intent is to improve spatially aware feature selection without a large complexity increase. The second refinement, **TransXNet-GG**, adds multiway dynamic dense connections (MUDD) in deeper stages to encourage short-range feature reuse. The **TransXNet-GGG** variant introduces differential attention in the later global branch. We denote the deployed main model as **UL-TransXNet**; in the ablation notation, it corresponds to the **TransXNet-GGG-MCA** configuration. This naming keeps the final model separate from the internal G/GG/GGG identifiers used for ablation. UL-TransXNet is used as the main cross-dataset model because the selection criterion is average behavior across TN5000, BUSI, and AUL rather than the TN5000-only optimum. The TN5000 ablations also show that the intermediate **TransXNet-GG** variant can be the strongest single TN5000 structure. This is an important part of the evidence rather than a contradiction: the refinements improve different operating characteristics, and MCA should be interpreted as a robustness-oriented module rather than as a guaranteed AUC-improving switch.

3.5. Key modules

Multi-scale coordinate attention (MCA).. When enabled, MCA operates on the global branch after channel splitting. It partitions channels into multiple scale groups, pools each group at a different spatial scale, injects coordinate-aware responses, and generates attention weights for spatially selective feature recalibration. In ultrasound images, lesions may differ through both coarse shape and fine echo patterns; MCA is intended to retain this scale diversity while keeping attention lightweight.

Multiway dynamic dense connection (MUDD).. MUDD is inserted in deeper stages, where semantic features are richer but optimization can be harder. The block feature is divided into multiple channel paths, and a lightweight weight generator computes path-wise aggregation weights from the current activation. Previous block features from the same stage are dynamically fused through these paths. This gives the deeper stages a compact adaptive memory without turning the backbone into a heavy dense network.

Differential attention.. The **GGG** variant replaces the standard global attention unit in selected later-stage locations with differential attention, within the general attention-based modeling framework introduced by Transformer architectures [26]. Queries and keys are split into two halves, the second query half is normalized, and two attention maps are formed and subtracted with a learnable balancing coefficient. The goal is to sharpen relational contrast in the global branch, which is useful when benign and malignant lesions share broad ultrasound appearance statistics but differ in subtle internal organization.

These modules are introduced as a progressive design family, not as a full factorial decomposition. The ablation study therefore reports both structure-level variants and core-module variants to make the interpretation explicit.

3.6. Complexity analysis

Table 1 reports the complexity of the comparison pool and the TransXNet family variants. The numbers are measured with the same binary-classification wrapper used by the training scripts, so they include both the backbone and the lightweight MLP head. All models use 256×256 ROI inputs except EfficientFormer-L1, which follows its 224×224 setting. UL-TransXNet is not the smallest model in the pool, but it remains substantially lighter than the strongest TN5000 baselines ConvNeXt-Tiny and Swin-T while preserving stronger cross-dataset behavior on BUSI and AUL.

4. Datasets and Experimental Setup

4.1. Datasets

The main paper uses three public ultrasound datasets: TN5000 for thyroid nodules [1], BUSI for breast lesions [2], and the Annotated Ultrasound Liver (AUL) dataset for liver lesions [3]. TN5000 is the largest benchmark and is used for the most detailed ablation analysis. BUSI and AUL are used to test whether the architecture remains competitive beyond thyroid ultrasound.

4.2. Implementation details

All datasets are processed in an ROI-based manner, but the crop expansion differs across benchmarks. TN5000 and BUSI use a lesion expansion

Table 1: Model complexity of the comparison pool and TransXNet family variants. Parameters and MACs are measured for the full binary classifier, including the lightweight MLP head. MACs are reported for one forward pass at the listed input size.

Method	Role	Input	Params (M)	MACs (G)
ResNet50	Prior baseline	256×256	24.56	5.40
DenseNet121	Prior baseline	256×256	7.48	3.70
EfficientNet-B0	Prior baseline	256×256	4.66	0.50
MobileNetV3-Large	Prior baseline	256×256	4.86	0.28
Swin-T	Prior baseline	256×256	27.91	7.11
ConvNeXt-Tiny	Prior baseline	256×256	28.21	5.82
RepViT-M1.1	Prior baseline	256×256	8.04	1.80
EfficientFormer-L1	Prior baseline	224×224	11.62	1.31
TransXNet	Ours / ablation	256×256	13.35	2.32
TransXNet-G	Ours / ablation	256×256	13.42	2.34
TransXNet-GG	Ours / ablation	256×256	14.15	2.38
TransXNet-GGG-noMCA	Ours / ablation	256×256	14.29	2.34
UL-TransXNet	Ours	256×256	14.37	2.36

ratio of 0.30, whereas AUL uses a lighter expansion ratio of 0.20. The lesion bounding box is read from XML or manifest metadata, expanded with a minimum crop size of 64 pixels, converted to a square crop, and resized to 256×256 ; when a reliable crop is unavailable, the whole image is used as a fallback. The only input-size exception in the comparison pool is EfficientFormer-L1, which is evaluated at 224×224 because its wrapper is not compatible with the 256×256 ROI setting used by the other models.

The training pipeline uses the same recipe family across the main runs:

Table 2: Overview of the main datasets used in the paper. BUSI and AUL are used in a benign–malignant lesion-classification setting.

Dataset	Organ	Images	B / M	Split protocol	Role
TN5000	Thyroid	5000	1435 / 3565	Official 3500/500/1000 train/val/test	Main
BUSI	Breast	647	435 / 212	Test 129; five folds on trainval 518	Cross-organ
AUL	Liver	635	183 / 452	Test 127; five folds on trainval 508	Cross-organ

random horizontal flip, random rotation ($\pm 4^\circ$), mild affine translation and scaling, color jitter on brightness and contrast, and random Gaussian blur. Evaluation uses deterministic resizing and ImageNet normalization only. All models use AdamW with weight decay 10^{-4} , a lightweight MLP head with dropout 0.3, and label smoothing 0.0. The learning-rate schedule is dataset-specific.

TN5000 uses the official train/val/test split and is reported with three random seeds (17, 27, and 37). The mainline setting uses 70 epochs, patience 12, manual class weights, and differential learning rates of 5×10^{-5} for the backbone and 1.5×10^{-4} for the classification head. BUSI uses the held-out test set together with five-fold training on the trainval subset; its main runs use 200 epochs, patience 36, class weighting, and the same $5 \times 10^{-5}/1.5 \times 10^{-4}$ learning-rate pair. AUL uses a held-out test set plus five folds on trainval, but uses a lighter budget: 20 epochs, patience 4, no class weighting, top-2 checkpoint ensembling, and lower differential learning rates of 1×10^{-5} for the backbone and 3×10^{-5} for the classification head. The mainline scripts enable exponential moving average (EMA) with decay 0.9995; the exception in the baseline pool is BUSI RepViT-M1.1, for which EMA is disabled because the EMA variant produced unstable folds during verification.

For inference-time calibration, each run uses validation-derived temperature scaling [24], averages the top- k validation-improved checkpoints (top-3 on TN5000 and BUSI, top-2 on AUL), applies horizontal-flip test-time augmentation, fits a scalar temperature on validation logits, and searches a decision threshold from 0.10 to 0.95 with step 0.01 using balanced accuracy as the default selection criterion. The resulting validation threshold is applied to the test predictions of that run. Main benchmark tables report mean $\pm$ standard deviation across the three TN5000 seeds or the five BUSI/AUL folds.

Importantly, all temperatures and decision thresholds are fixed strictly from validation-derived predictions. No test labels are used for temperature fitting, threshold tuning, or operating-point selection; the test set is used only once for final reporting after these choices are fixed.

4.3. Automatic ROI closed-loop protocol

The main experiments use annotation-derived ROI crops to isolate classifier behavior under a controlled input protocol. To evaluate whether this ROI classifier can be connected to an automatic front end at inference time, we additionally train one-class lesion detectors on BUSI and AUL and perform closed-loop classification with detector-predicted crops. The detector is used only as an auxiliary ROI generator; it is not presented as a novel detection method.

For each BUSI/AUL fold, a one-class YOLO detector is trained on the corresponding training annotations and evaluated on the held-out test images. The predicted lesion box with the highest confidence is exported into a VOC-style annotation file and passed through the same expansion, square-

crop, resize, normalization, temperature-scaling, and thresholding code used by the ROI classifier. We then compare three test-time inputs using the same trained UL-TransXNet fold checkpoints: *oracle ROI*, which uses the ground-truth lesion annotation; *automatic ROI*, which uses the detector-predicted box; and *full image*, which disables the lesion crop and resizes the complete ultrasound image. Thus, the closed-loop experiment tests whether detector-predicted crops can partially substitute annotation-derived ROI construction without retraining the classifier.

4.4. Compared methods

The benchmark pool combines representative CNN, lightweight hybrid, and modern efficient baselines. The comparison set includes ResNet50 [6], EfficientNet-B0 [8], MobileNetV3-Large [9], Swin-T [12], DenseNet121 [7], ConvNeXt-Tiny [13], RepViT-M1.1 [17], and EfficientFormer-L1 [16]. The main paper table reports *Ours* together with the strongest non-*Ours* baseline on each dataset, where “strongest” is determined by mean test AUC under the same reporting convention. The full baseline table is provided in Appendix Table A.14.

4.5. Evaluation metrics and statistical analysis

The main metrics are AUC, balanced accuracy (BalAcc), macro F1, and accuracy. Unless explicitly stated otherwise, “F1” in the manuscript refers to macro F1. Because the tables emphasize repeated-seed/fold stability, results are reported as mean $\pm$ standard deviation. Thresholded metrics should be interpreted as validation-selected operating points rather than test-tuned oracle points.

4.6. Local efficiency profiling

Model efficiency is profiled on the TN5000 test split, which provides the largest held-out set among the main datasets. Local CUDA latency is measured on an NVIDIA GeForce RTX 4080 SUPER using batch size 1. Each model is instantiated with the same binary-classification wrapper used for complexity profiling, warmed up for 50 forward passes, and then timed for 300 model-only forward passes with CUDA synchronization. We report the mean model-only latency and the corresponding FPS. A separate TN5000 pipeline pass over 1000 test ROI crops is also logged for diagnosis, but the main trade-off table uses model-only latency to avoid mixing architecture cost with image decoding and CPU preprocessing overhead.

4.7. Mobile deployment protocol

Mobile deployment is evaluated with an Android prototype built on ONNX Runtime 1.24.3. The reported batch experiments were executed on a Xiaomi device (model 24129PN74C, Android 16, arm64-v8a). The deployment study uses TN5000-calibrated UL-TransXNet ONNX exports and compares three inference variants: a single checkpoint, the same checkpoint with horizontal-flip test-time augmentation (TTA), and a three-checkpoint ensemble with TTA. Cold-start runs explicitly cleared the session cache before evaluation, whereas warm-start runs reused the loaded model session. The single-checkpoint mode was measured with one cold-start and three warm-start runs; the TTA and ensemble variants were measured with one cold-start and one warm-start run each. The deployment configuration used lesion expansion ratio 0.30, a fitted temperature of 1.819830, and decision threshold 0.73 inherited from TN5000 validation calibration. We report

Table 3: Main comparison results on the public ultrasound datasets. For each dataset, the table reports the proposed model and the strongest non-*Ours* baseline ranked by mean test AUC. F1 denotes macro F1.

Dataset	Method	AUC	BalAcc	F1-macro	Acc
TN5000	UL-TransXNet	0.9483 ± 0.0006	0.8726 ± 0.0046	0.8503 ± 0.0123	0.8790 ± 0.0140
TN5000	ConvNeXt-Tiny	0.9508 ± 0.0020	0.8794 ± 0.0023	0.8731 ± 0.0066	0.8957 ± 0.0065
BUSI	UL-TransXNet	0.8998 ± 0.0116	0.7961 ± 0.0104	0.8018 ± 0.0119	0.8140 ± 0.0130
BUSI	Swin-T	0.8190 ± 0.0091	0.7137 ± 0.0095	0.6896 ± 0.0300	0.7194 ± 0.0445
AUL	UL-TransXNet	0.8767 ± 0.0009	0.8200 ± 0.0487	0.7462 ± 0.0405	0.7685 ± 0.0418
AUL	Swin-T	0.8319 ± 0.0117	0.7783 ± 0.0325	0.7589 ± 0.0243	0.7717 ± 0.0211

average end-to-end latency and the corresponding 95th percentile latency over the full ROI preprocessing-plus-inference pipeline.

5. Results

5.1. Main comparison results on public ultrasound datasets

Table 3 summarizes the main comparison results, reported as mean $\pm$ standard deviation across TN5000 seeds or BUSI/AUL folds. The evidence should be read carefully. On TN5000, ConvNeXt-Tiny is slightly stronger than the proposed model at the selected operating point. On BUSI and AUL, however, UL-TransXNet provides a clear gain over the strongest non-*Ours* baseline. Averaged over the three datasets, the proposed model gives the strongest overall performance, especially in balanced accuracy.

These results delimit the central claim. The proposed model should not be described as uniformly better than every baseline on every dataset. Its advantage is cross-dataset: it is competitive on TN5000 and substantially stronger on BUSI and AUL. The mean AUC over TN5000, BUSI, and AUL

is 0.9083 for UL-TransXNet, compared with 0.8672 for the strongest-per-dataset non-*Ours* rows listed in Table 3. The corresponding mean balanced accuracy is 0.8296 versus 0.7905. This supports the claim that the architecture is a strong lightweight multi-dataset ultrasound classifier, not a claim of uniform dominance on the primary thyroid benchmark.

Table 4 adds an offline paired statistical check over the same repeated runs used by Table 3. On TN5000, the proposed model is slightly below ConvNeXt-Tiny. On BUSI and AUL, the paired differences are consistently positive for both AUC and balanced accuracy. Because the available repeated units are only three TN5000 seeds and five BUSI/AUL folds, these tests are used as stability evidence rather than as large-sample clinical significance claims.

Table 5 gives the corresponding complexity–performance view for the full comparison pool. This table makes the deployment trade-off more explicit than the accuracy-only table: UL-TransXNet is not the smallest or fastest model, but it achieves the highest mean AUC and mean balanced accuracy with 14.37M parameters and 2.36 GMACs. Its local CUDA batch-1 latency is 30.23 ms, which is slower than the most GPU-optimized baselines but remains within an interactive range. In contrast, the strongest TN5000 baselines ConvNeXt-Tiny and Swin-T require 28.21M/5.82G and 27.91M/7.11G, respectively. The input column is included because EfficientFormer-L1 is evaluated at its native 224 input while the other models use the 256 ROI input used by the proposed model.

Table 4: Offline paired statistical comparison between UL-TransXNet and the strongest non-*Ours* baseline in Table 3. Δ denotes UL-TransXNet minus baseline. Confidence intervals are paired bootstrap 95% intervals over seeds or folds; p is an exact two-sided paired sign-flip test. Because only three TN5000 seeds and five BUSI/AUL folds are available, these tests are intended as repeated-run stability checks rather than large-sample clinical significance tests.

Dataset	Baseline	Metric	n	Δ	95% CI	p
TN5000	ConvNeXt-Tiny	AUC	3	-0.0025	[-0.0040, -0.0006]	0.2500
TN5000	ConvNeXt-Tiny	BalAcc	3	-0.0068	[-0.0104, -0.0022]	0.2500
BUSI	Swin-T	AUC	5	+0.0807	[0.0697, 0.0918]	0.0625
BUSI	Swin-T	BalAcc	5	+0.0823	[0.0647, 0.0961]	0.0625
AUL	Swin-T	AUC	5	+0.0449	[0.0375, 0.0574]	0.0625
AUL	Swin-T	BalAcc	5	+0.0417	[0.0160, 0.0674]	0.0625

5.2. TN5000 structure ablation

Table 6 reports the unified-protocol TN5000 structure ablation. The strongest TN5000 structure is **TransXNet-GG**, which achieves the highest AUC, balanced accuracy, macro F1, and accuracy in this ablation group. The **TransXNet-GGG-MCA** ablation row, which corresponds to UL-TransXNet, is close but not best on TN5000 alone. This distinction is important because it prevents the paper from over-attributing all gains to the final GGG-MCA setting.

The structure ablation shows that the TransXNet-family modifications are useful but metric-dependent. The G and GG variants improve thresholded operating-point behavior relative to the baseline. The GGG-MCA con-

Table 5: Complexity–performance trade-off of the comparison pool. CUDA latency is the mean model-only batch-1 latency measured on the local workstation; AUC values are mean test AUCs across TN5000 seeds or BUSI/AUL folds. Mean AUC and Mean BalAcc are averaged over the three datasets.

Method	Type	Input	Params (M)↓	MACs (G)↓	CUDA ms↓	CUDA FPS↑	AUC↑			Mean AUC↑	Mean BalAcc↑
							TN5000	BUSI	AUL		
MobileNetV3-Large	Mobile CNN	256×256	4.86	0.28	3.17	315.1	0.8282	0.7947	0.7357	0.7862	0.7183
EfficientNet-B0	Efficient CNN	256×256	4.66	0.50	6.53	153.1	0.7856	0.7659	0.6099	0.7205	0.6682
EfficientFormer-L1	Hybrid	224×224	11.62	1.31	3.41	293.0	0.9249	0.7917	0.7196	0.8121	0.7330
RepViT-M1.1	Mobile CNN	256×256	8.04	1.80	5.19	192.5	0.8403	0.8028	0.6797	0.7743	0.7020
UL-TransXNet	Ours	256×256	14.37	2.36	30.23	33.1	0.9483	0.8998	0.8767	0.9083	0.8296
DenseNet121	CNN	256×256	7.48	3.70	7.19	139.1	0.9360	0.8154	0.6729	0.8081	0.7401
ResNet50	CNN	256×256	24.56	5.40	3.25	307.5	0.6531	0.7519	0.5481	0.6510	0.6190
ConvNeXt-Tiny	ConvNet	256×256	28.21	5.82	3.30	303.3	0.9508	0.8050	0.8274	0.8611	0.7651
Swin-T	Transformer	256×256	27.91	7.11	6.32	158.3	0.9497	0.8190	0.8319	0.8669	0.7901

Table 6: Unified-protocol structure ablation on TN5000. Mean and standard deviation are computed over three seeds.

Variant	AUC	BalAcc	F1-macro	Acc
TransXNet	0.9495 ± 0.0020	0.8697 ± 0.0100	0.8366 ± 0.0222	0.8640 ± 0.0230
TransXNet-G	0.9481 ± 0.0013	0.8745 ± 0.0074	0.8395 ± 0.0220	0.8660 ± 0.0233
TransXNet-GG	0.9517 ± 0.0011	0.8784 ± 0.0076	0.8552 ± 0.0173	0.8827 ± 0.0170
TransXNet-GGG-noMCA	0.9503 ± 0.0019	0.8660 ± 0.0069	0.8325 ± 0.0013	0.8610 ± 0.0016
TransXNet-GGG-MCA	0.9483 ± 0.0006	0.8726 ± 0.0046	0.8503 ± 0.0123	0.8790 ± 0.0140

figuration is not the TN5000 AUC winner, but it remains competitive and corresponds to the model that generalizes best across the three main datasets. Table 7 makes this selection criterion explicit: **TransXNet-GG** is the strongest TN5000-only structure, whereas UL-TransXNet is substantially stronger on BUSI and AUL and therefore gives the better three-dataset average.

Table 7: TN5000-only optimum versus the final cross-dataset configuration. The final model is selected for three-dataset behavior, not for the single best TN5000 ablation row.

Dataset	TransXNet-GG		UL-TransXNet	
	AUC	BalAcc	AUC	BalAcc
TN5000	0.9517	0.8784	0.9483	0.8726
BUSI	0.7707	0.7000	0.8998	0.7961
AUL	0.6972	0.6550	0.8767	0.8200
Mean	0.8065	0.7445	0.9083	0.8296

5.3. TN5000 core-module ablation

Table 8 isolates the core-module path under the same TN5000 protocol. Differential attention provides the clearest single-module operating-point improvement: **GGG-core+DA** improves balanced accuracy, macro F1, and accuracy over **GGG-core**. MUDD alone gives a smaller gain in thresholded metrics and does not improve AUC in this setting. Adding MCA to the full GGG configuration lowers TN5000 AUC slightly relative to the no-MCA full model, but improves balanced accuracy, macro F1, and accuracy. This supports a restrained interpretation of MCA: it is useful for decision-level robustness and cross-dataset behavior, not a universal AUC booster.

5.4. MCA effect across datasets

Table 9 compares the final GGG model with and without MCA across the three main datasets. The result explains why the final manuscript should keep MCA in the model family but avoid overclaiming it. On TN5000, MCA improves balanced accuracy, macro F1, and accuracy while slightly reducing

Table 8: Unified-protocol core-module ablation on TN5000. Mean and standard deviation are computed over three seeds.

Variant	AUC	BalAcc	F1-macro	Acc
GGG-core	0.9496 ± 0.0003	0.8704 ± 0.0083	0.8324 ± 0.0136	0.8593 ± 0.0133
GGG-core+MUDD	0.9470 ± 0.0006	0.8712 ± 0.0045	0.8406 ± 0.0097	0.8687 ± 0.0111
GGG-core+DA	0.9489 ± 0.0016	0.8751 ± 0.0044	0.8490 ± 0.0100	0.8770 ± 0.0099
GGG-core+MUDD+DA-noMCA	0.9503 ± 0.0019	0.8660 ± 0.0069	0.8325 ± 0.0013	0.8610 ± 0.0016
GGG-core+MUDD+DA+MCA	0.9483 ± 0.0006	0.8726 ± 0.0046	0.8503 ± 0.0123	0.8790 ± 0.0140

Table 9: Effect of enabling MCA in the final GGG configuration. Mean and standard deviation are computed over TN5000 seeds or BUSI/AUL folds.

Dataset	Variant	AUC	BalAcc	F1-macro	Acc
TN5000	GGG-noMCA	0.9503 ± 0.0019	0.8660 ± 0.0069	0.8325 ± 0.0013	0.8610 ± 0.0016
TN5000	GGG-MCA	0.9483 ± 0.0006	0.8726 ± 0.0046	0.8503 ± 0.0123	0.8790 ± 0.0140
BUSI	GGG-noMCA	0.8588 ± 0.0137	0.7551 ± 0.0295	0.7526 ± 0.0300	0.7628 ± 0.0288
BUSI	GGG-MCA	0.8998 ± 0.0116	0.7961 ± 0.0104	0.8018 ± 0.0119	0.8140 ± 0.0130
AUL	GGG-noMCA	0.8720 ± 0.0046	0.8181 ± 0.0338	0.7462 ± 0.0283	0.7685 ± 0.0321
AUL	GGG-MCA	0.8767 ± 0.0009	0.8200 ± 0.0487	0.7462 ± 0.0405	0.7685 ± 0.0418

AUC. On BUSI, MCA provides a large gain across all reported metrics. On AUL, MCA gives a small AUC and balanced-accuracy gain with nearly unchanged macro F1 and accuracy. Thus, MCA is best described as a module that improves cross-dataset operating behavior, especially on BUSI, rather than as a universally dominant change.

5.5. Automatic ROI localization and closed-loop classification

The main ROI protocol isolates classifier behavior using annotation-derived crops. Figure 4, Table 10, and Table 11 evaluate a more complete inference chain: an automatic detector first predicts the lesion box, an

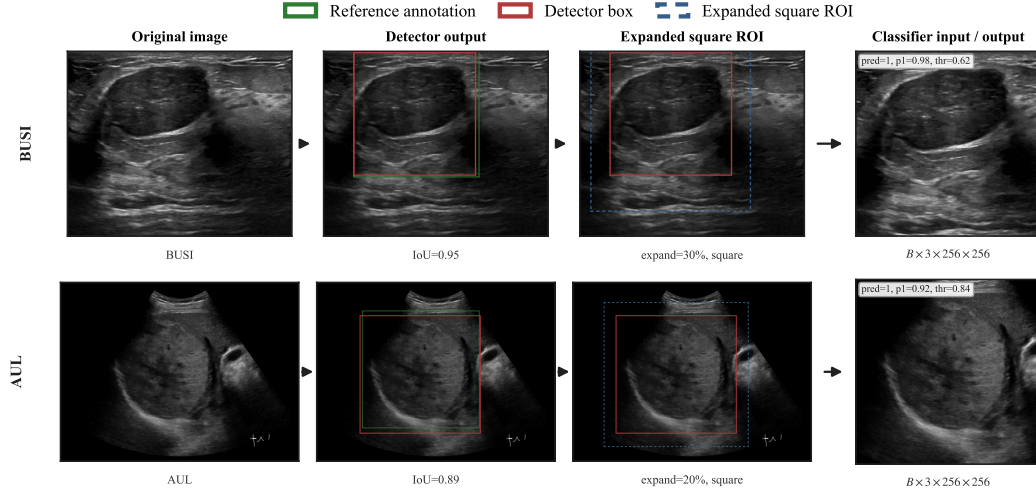


Figure 4: Automatic ROI inference workflow for the BUSI/AUL closed-loop evaluation. Green boxes indicate reference annotations used only for evaluation and visualization; red boxes are detector-predicted lesion boxes; blue dashed boxes are the expanded square ROIs used as classifier inputs.

expanded square ROI is constructed from that prediction, and the same trained UL-TransXNet classifiers are then applied without retraining. The detector does not miss any BUSI or AUL test image in this evaluation, but localization quality differs by dataset. BUSI reaches a mean IoU of 0.7365, whereas AUL reaches 0.5755 and has a lower high-IoU recall.

Automatic ROI classification is consistently stronger than full-image classification. On BUSI, the automatic ROI improves AUC by 0.0988 and balanced accuracy by 0.0857 over the full-image input. On AUL, the corresponding gains are 0.1066 AUC and 0.0789 balanced accuracy. The automatic ROI remains below oracle ROI, especially on AUL, which is consistent with the lower localization quality in Table 10. These results support the practical use of an automatic ROI front end, while keeping the detector itself outside

Table 10: Detector localization quality used for automatic ROI construction. Values are mean $\pm$ standard deviation across five detector folds.

Dataset	Mean IoU	R@0.50	R@0.75	No det.
BUSI	0.7365 $\pm$ 0.0241	0.8403 $\pm$ 0.0141	0.6822 $\pm$ 0.0548	0.0000
AUL	0.5755 $\pm$ 0.0221	0.6740 $\pm$ 0.0419	0.4331 $\pm$ 0.0201	0.0000

Table 11: Closed-loop classification with oracle ROI, automatic detector ROI, and full-image input. Values are mean $\pm$ standard deviation across five folds. Bracketed intervals are fold-bootstrap 95% confidence intervals for AUC and balanced accuracy.

Dataset	Input	AUC	BalAcc	F1-macro	Acc
BUSI	oracle	0.8998 $\pm$ 0.0116 [0.8880, 0.9067]	0.7961 $\pm$ 0.0104 [0.7865, 0.8045]	0.8018 $\pm$ 0.0119	0.8140 $\pm$ 0.0130
BUSI	auto	0.8725 $\pm$ 0.0158 [0.8564, 0.8845]	0.7839 $\pm$ 0.0179 [0.7704, 0.8010]	0.7893 $\pm$ 0.0176	0.8031 $\pm$ 0.0167
BUSI	full	0.7737 $\pm$ 0.0044 [0.7701, 0.7777]	0.6982 $\pm$ 0.0237 [0.6762, 0.7178]	0.6915 $\pm$ 0.0378	0.7023 $\pm$ 0.0418
AUL	oracle	0.8767 $\pm$ 0.0009 [0.8759, 0.8776]	0.8200 $\pm$ 0.0487 [0.7755, 0.8599]	0.7462 $\pm$ 0.0405	0.7685 $\pm$ 0.0418
AUL	auto	0.8507 $\pm$ 0.0222 [0.8320, 0.8708]	0.7593 $\pm$ 0.0288 [0.7340, 0.7847]	0.6908 $\pm$ 0.0561	0.7165 $\pm$ 0.0713
AUL	full	0.7441 $\pm$ 0.0111 [0.7344, 0.7538]	0.6805 $\pm$ 0.0214 [0.6623, 0.6999]	0.6787 $\pm$ 0.0203	0.7512 $\pm$ 0.0190

the main architectural contribution.

5.6. Qualitative Grad-CAM sanity check

Figure 5 provides a qualitative Grad-CAM sanity check using one trained UL-TransXNet checkpoint per dataset. The maps are generated from late spatial backbone blocks selected for readable ROI-level maps. The activations broadly fall on lesion tissue and adjacent ultrasound context inside the ROI. This visualization is used only as a qualitative sanity check that the classifier is not driven solely by image borders or unrelated background. It is not used as quantitative evidence of causal attention or as a substitute for the ablation and closed-loop ROI experiments above.

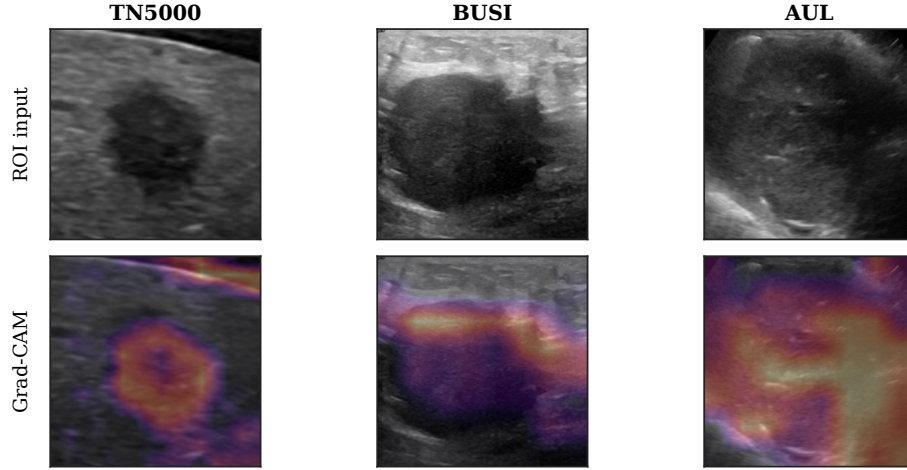


Figure 5: Grad-CAM sanity-check visualization for representative correctly classified ROI inputs. Columns correspond to datasets; the upper row shows the ROI input and the lower row shows the Grad-CAM overlay. The maps are generated from a single UL-TransXNet checkpoint for each dataset and are intended as qualitative localization checks only.

5.7. Uncertainty and calibration analysis

Table 12 reports bootstrap confidence intervals and calibration metrics for UL-TransXNet using the same seed/fold-level reporting protocol as the main tables. The confidence intervals provide a compact statistical view of the repeated-run stability, while ECE and Brier score quantify probability calibration after the validation-derived temperature-scaling step.

5.8. Efficiency and mobile deployment

The Android deployment experiment evaluates a TN5000-calibrated prototype of UL-TransXNet. Table 13 and Figure 6 show a clear deployment trade-off among three inference variants. A single checkpoint provides the lowest end-to-end latency at 116.5 ms per image. Adding horizontal-flip TTA improves both AUC and balanced accuracy, reaching 0.9466 AUC and 0.8762

Table 12: Uncertainty and calibration summary for UL-TransXNet on the main test sets. Values follow the manuscript seed/fold-level reporting protocol; intervals are bootstrap 95% confidence intervals over seeds or folds. ECE uses 10 confidence bins.

Dataset	AUC 95% CI	BalAcc 95% CI	ECE	Brier
TN5000	0.9483 [0.9475, 0.9488]	0.8726 [0.8662, 0.8771]	0.0220	0.0741
BUSI	0.8998 [0.8880, 0.9067]	0.7961 [0.7865, 0.8045]	0.0696	0.1358
AUL	0.8767 [0.8759, 0.8776]	0.8200 [0.7755, 0.8599]	0.1283	0.1597

Table 13: Android deployment results on the TN5000 test set. Latency includes ROI preprocessing and ONNX Runtime inference.

Mode	Runs	AUC	BalAcc	Avg ms	P95 ms
Single	4	0.9402	0.8688	116.5	129.0
+ HFlip TTA	2	0.9466	0.8762	226.2	246.6
3-ckpt ensemble + TTA	2	0.9485	0.8723	738.6	763.5

balanced accuracy at 226.2 ms per image. The three-checkpoint ensemble further increases AUC to 0.9485, but it reduces balanced accuracy at the selected operating threshold and raises latency to 738.6 ms per image. Thus, the single and TTA variants are the practical interactive settings, whereas the ensemble is better interpreted as an offline ranking or review option.

6. Discussion

The results support a restrained conclusion. UL-TransXNet is a competitive lightweight ultrasound classifier with strong cross-dataset behavior, but it should not be described as uniformly superior on every dataset and ev-

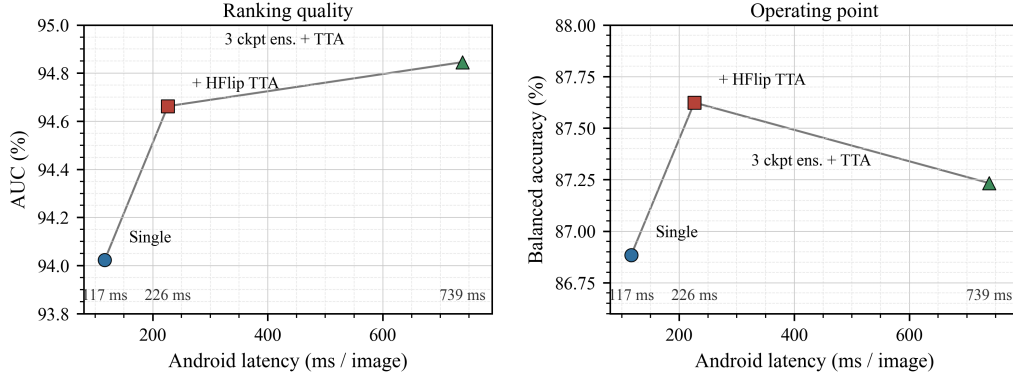


Figure 6: Accuracy-latency trade-off of the Android deployment variants on the TN5000 test set.

ery metric. On TN5000, ConvNeXt-Tiny remains slightly stronger in the repeated-seed table. The proposed model becomes clearly advantageous when BUSI and AUL are included: it is substantially stronger than the strongest non-*Ours* baselines on both cross-organ datasets and has the best average performance across the three public benchmarks.

The TN5000 ablations clarify the role of the architecture components. The structure-level progression shows that **TransXNet-GG** is the strongest TN5000 variant, while the **TransXNet-GGG-MCA** configuration used by UL-TransXNet is the best cross-dataset choice. The module-level ablation shows that differential attention is the clearest contributor to TN5000 thresholded metrics. MUDD has a smaller and less uniform effect. MCA does not uniformly improve TN5000 AUC, but it improves TN5000 F1 and accuracy and produces large BUSI gains. This means that MCA should be framed as a robustness and operating-point module rather than as the sole reason for the model family performance.

This interpretation is important in medical image analysis. Ultrasound datasets differ substantially in organ anatomy, acquisition quality, lesion size, and class prevalence. A small AUC difference on one thyroid benchmark is less informative than the pattern across multiple organs and operating points. The evidence suggests that the proposed architecture is useful because it gives a favorable balance across datasets, not because every module monotonically improves every metric.

The automatic ROI experiment addresses a separate practical concern. The main classifier is trained under an annotation-derived ROI protocol, but the closed-loop BUSI/AUL evaluation shows that detector-predicted crops can replace oracle crops with a moderate performance loss and remain clearly stronger than full-image classification. This result should be interpreted as workflow evidence rather than detector novelty. In particular, AUL localization is weaker than BUSI localization, which explains part of the remaining gap between automatic and oracle ROI classification.

The Android results remain a practical strength of the study. Many ultrasound classification papers stop at workstation-side accuracy. The mobile experiment shows that a TN5000-calibrated prototype can run in a low-latency mode while preserving strong classification quality, and that a more expensive ensemble mode is available when latency is less critical. These measurements do not constitute clinical deployment validation, but they show that UL-TransXNet is compatible with realistic edge-side inference constraints.

Limitations

This study has several limitations. First, the evaluation is still based on public retrospective datasets, and label definitions, class proportions, and

preprocessing conditions are not fully uniform across datasets. Second, the work focuses on binary benign–malignant classification rather than detection, segmentation, or multi-class clinical diagnosis. Third, the automatic ROI experiment uses a practical one-class detector only to form the classifier input; it does not establish a new lesion-detection method, and the AUL results show that localization quality can still limit closed-loop performance. Fourth, the mobile experiment demonstrates feasibility but not a full clinical workflow under realistic operator interaction and acquisition variability. Finally, external robustness remains an open problem because the evaluation is still limited to public retrospective datasets rather than prospective multi-center cohorts. Future studies should therefore focus on broader external validation, stronger detector-classifier integration, calibration under distribution shift, and clinically grounded prospective evaluation.

7. Conclusion

This paper presents UL-TransXNet, a lightweight ROI-based TransXNet-family architecture for benign–malignant ultrasound lesion classification. The evidence supports a restrained but coherent conclusion. UL-TransXNet is competitive on TN5000, clearly stronger on BUSI and AUL, and strongest on average across the three public datasets under the unified reporting protocol. TN5000 ablations show that the architecture gains are not explained by a single module: the GG structure is the strongest thyroid-only variant, differential attention improves thresholded behavior, and MCA mainly improves cross-dataset operating robustness. The automatic ROI experiment further shows that detector-predicted crops can connect the classifier to a

Table A.14: Full benchmark across TN5000, BUSI, and AUL using the unified reporting convention. Best per column is shown in bold.

Method	TN5000 AUC	TN5000 BalAcc	BUSI AUC	BUSI BalAcc	AUL AUC	AUL BalAcc
UL-TransXNet	0.9483 $\pm$ 0.0006	0.8726 $\pm$ 0.0046	0.8998 $\pm$ 0.0116	0.7961 $\pm$ 0.0104	0.8767 $\pm$ 0.0009	0.8200 $\pm$ 0.0487
ResNet50	0.6531 $\pm$ 0.0081	0.6319 $\pm$ 0.0067	0.7519 $\pm$ 0.0332	0.6826 $\pm$ 0.0205	0.5481 $\pm$ 0.0771	0.5426 $\pm$ 0.0343
EfficientNet-B0	0.7856 $\pm$ 0.0242	0.7026 $\pm$ 0.0165	0.7659 $\pm$ 0.0368	0.7061 $\pm$ 0.0317	0.6099 $\pm$ 0.1220	0.5958 $\pm$ 0.0912
MobileNetV3-Large	0.8282 $\pm$ 0.0083	0.7626 $\pm$ 0.0082	0.7947 $\pm$ 0.0547	0.7311 $\pm$ 0.0462	0.7357 $\pm$ 0.0062	0.6611 $\pm$ 0.0161
Swin-T	0.9497 $\pm$ 0.0025	0.8782 $\pm$ 0.0051	0.8190 $\pm$ 0.0091	0.7137 $\pm$ 0.0095	0.8319 $\pm$ 0.0117	0.7783 $\pm$ 0.0325
DenseNet121	0.9360 $\pm$ 0.0062	0.8609 $\pm$ 0.0074	0.8154 $\pm$ 0.0278	0.7291 $\pm$ 0.0471	0.6729 $\pm$ 0.0358	0.6304 $\pm$ 0.0706
ConvNeXt-Tiny	0.9508 $\pm$ 0.0020	0.8794 $\pm$ 0.0023	0.8050 $\pm$ 0.0237	0.7261 $\pm$ 0.0420	0.8274 $\pm$ 0.0284	0.6898 $\pm$ 0.0248
RepViT-M1.1	0.8403 $\pm$ 0.0510	0.7603 $\pm$ 0.0427	0.8028 $\pm$ 0.0134	0.7137 $\pm$ 0.0442	0.6797 $\pm$ 0.0029	0.6320 $\pm$ 0.0225
EfficientFormer-L1	0.9249 $\pm$ 0.0021	0.8484 $\pm$ 0.0088	0.7917 $\pm$ 0.0342	0.7213 $\pm$ 0.0226	0.7196 $\pm$ 0.0136	0.6293 $\pm$ 0.0544

practical localization front end and remain clearly stronger than full-image classification on BUSI and AUL. Android experiments show that the model can support practical speed–accuracy trade-offs on mobile hardware. At the same time, broad external generalization remains unresolved because all reported datasets are public retrospective benchmarks. Future studies should therefore focus on larger external validation, stronger automatic ROI generation, more rigorous calibration under distribution shift, and deployment evaluation under realistic clinical workflows.

Appendix A. Supplementary Benchmark Tables

Table A.14 reports the full benchmark table used to support the main comparison. Values are mean $\pm$ standard deviation across TN5000 seeds or BUSI/AUL folds. The proposed model is UL-TransXNet, which corresponds to the TransXNet-GGG-MCA configuration in the ablation notation; baseline rows are taken from the same comparison protocol.

Appendix B. Reproducibility Note

All main benchmark tables were generated from the seed/fold summary CSV files used by the experiment pipeline. The project archive records the dataset split manifests, ROI construction scripts, training entry points, aggregation scripts, model checkpoints, Android deployment artifacts, and a table-level source index required to reproduce the reported TN5000, BUSI, AUL, and Android deployment results.

CRedit authorship contribution statement

Zihan Li: Conceptualization, methodology, software, experiments, and writing—original draft.

Cong Yang: Supervision, conceptualization, writing—review and editing, resources, and project administration.

Ethics statement

This study is a secondary analysis of publicly available, de-identified ultrasound datasets. No new patient data were collected by the authors. Ethical approval and consent procedures are governed by the original dataset publications or data providers; therefore, no additional institutional review board approval was required for this retrospective computational study.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed in this study are publicly available from their original sources, which are cited in the manuscript. The source code, paper-generation scripts, Android demonstration source, and compact result summaries are available at <https://github.com/Afr1ste/ul-transxnet>. Dataset files, trained checkpoints, ONNX binaries, and Android application packages are not redistributed in the repository; they are archived with the project materials and are available from the corresponding author upon reasonable request, subject to the applicable redistribution terms of the original datasets.

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